

Chapter 7: Decimals

11 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.

CHAPTER

7

7.01 Ordering decimals

7.02 Decimals and fractions

7.03 Adding and subtracting decimals

7.04 Multiplying and dividing by powers of 10

7.05 Using a known product

7.06 Multiplying decimals

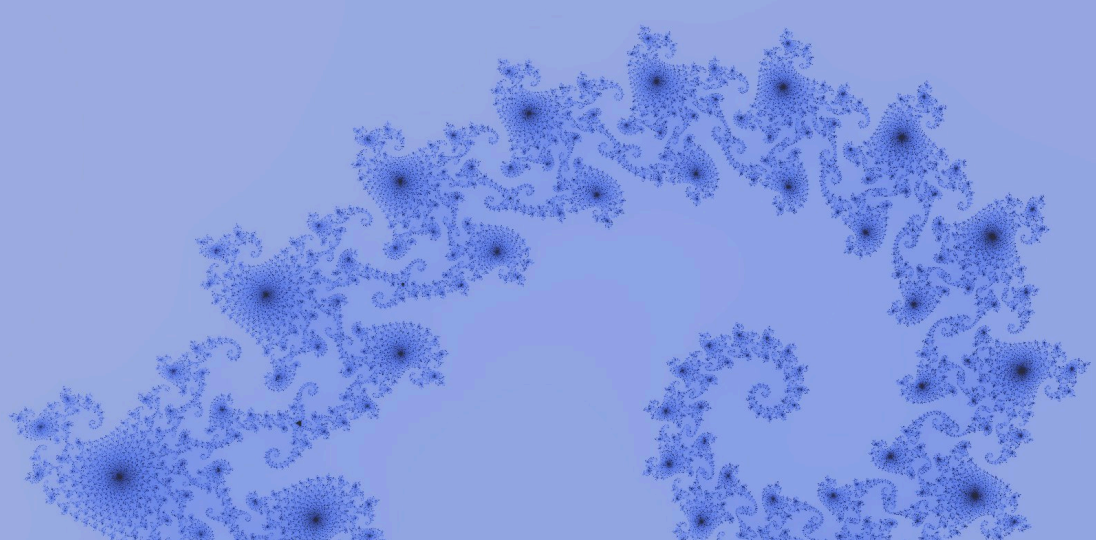
7.07 Dividing decimals by whole numbers

7.08 Dividing by a decimal

7.09 Fractions to decimals

7.10 Rounding decimals

7.11 Decimals and money



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

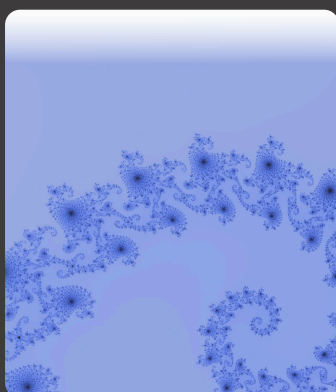
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
7.01	Ordering decimals	3		☐
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In this chapter you will

- order decimals
- convert between decimals and fractions
- add, subtract, multiply and divide decimals
- multiply and divide by powers of 10
- round decimals and write recurring decimals
- solve money problems



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a double spiral deep inside Seahorse Valley. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
7.01

Ordering decimals

Date _____

Learning intentions

We are learning to:

- compare and order decimals

Success criteria

I can:

- ☐ use place value to compare decimals
- ☐ fill with zeros so decimals have the same number of places
- ☐ arrange decimals in ascending and descending order

Key terms Write the definition as your teacher explains it.

Ascending

Descending

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Which is larger?

a Which is larger: 0.7 or 0.65?

b Which is larger: 3.998 or 34.89?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Line up the decimal points.

ROUND 1 Questions 1–16: which decimal is larger?

1 8.37 or 8.42

2 1.69 or 1.64

3 2.7 or 2.63

4 8.6 or 8.8

5 3.92 or 3.84

6 9.55 or 9.64

7 3.33 or 3.39

8 7.72 or 8.22

9 6.91 or 6.82

10 8.52 or 8.55

11 1.19 or 1.9

12 2.8 or 3.5

13 5.58 or 5.6

14 4.71 or 4.1

15 9.12 or 9.15

16 2.4 or 3.2

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: write the value of the digit in bold.

17 3.89**2**

18 5.**7**56

19 5.**3**11

20 2.12**9**

21 6.49**8**

22 **8**.457

23 1.7**9**3

24 **5**.854

25 9.9**5**4

26 9.98**1**

27 5.**8**25

28 9.39**3**

29 **7**.598

30 8.77**2**

31 4.78**4**

32 8.34**2**



Level 2 · Building up

Medium · 16 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write the place value, then order the decimals.

a Write the place value of 8 in 12.084.

Your teacher works this out. Copy every step.

b Arrange 34.98, 36.86, 3.998, 38.141, 340, 34.89 in ascending order.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

REMEMBER Fill empty places with zeros.

ROUND 1

Questions 1–8: write the decimals in ascending order (smallest first).

1 1.34, 1.579, 1.85, 1.4

2 2.8, 2.15, 2.685, 2.89

3 3.8, 3.72, 3.965, 3.64

4 5.52, 5.96, 5.38, 5.1

5 1.9, 1.68, 1.824, 1.75

6 2.98, 2.672, 2.53, 2.5

7 3.81, 3.578, 3.51, 3.9

8 0.93, 0.897, 0.96, 0.8

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 9–16: write < or > in the box.

9 $4.685 \square 2.67$

10 $0.579 \square 1.47$

11 $7.699 \square 3.21$

12 $5.056 \square 8.96$

13 $9.389 \square 5.55$

14 $7.09 \square 0.83$

15 $0.795 \square 0.77$

16 $1.179 \square 1.94$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order the decimals.

a Arrange 1.015, 1.293, 1.1015, 1.239, 1.006 in descending order.

Your teacher works this out. Copy every step.

b Write a decimal between 2.35 and 2.36.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Order and compare.

3 QUESTIONS

REMEMBER Compare from the left, place by place.

1 Arrange 1.015, 1.293, 1.1015, 1.239, 1.006 in descending order.

2 Write a decimal between 2.35 and 2.36.

3 Four runners: 12.08 s, 12.8 s, 12.018 s, 12.108 s. Who won?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Which is larger: 0.4 or 0.38?

Level 2

Arrange 5.2, 5.02, 5.22 in ascending order.

Level 3

Arrange 0.3, 0.303, 0.033, 0.33 in descending order.

Extension · Stretch yourself

Worked example

Write the decimal halfway between 3.47 and 3.5.

1. $3.5 = 3.50$
2. Halfway between 3.47 and 3.50 is 3.485

Answer: 3.485

YOU DO

On your own: read the worked example, then try these.

- E1** Find the decimal halfway between 0.6 and 0.61.

- E2** How many decimals with 2 decimal places are between 4.3 and 4.4?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Line up the decimal points.
 2. Fill empty places with zeros.
 3. Compare from the left, place by place.
 4. Ascending: smallest first. Descending: largest first.

Level 1

0.7 or 0.65

1. 0.70 vs 0.65

Answer: 0.7

Level 2

Place of 8 in 12.084

1. Hundredths

Answer: 8 hundredths

Level 3

Ascending: 34.98, 3.998, 340...

1. Compare whole numbers first

Answer: 3.998, 34.89, 34.98, 36.86, 38.141, 340

Extension

Halfway 3.47 and 3.5

1. 3.470 and 3.500

Answer: 3.485

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.02

Decimals and fractions

Date _____

Learning intentions

We are learning to:

- convert between decimals and fractions

Success criteria

I can:

- ☐ write a decimal as a fraction over 10, 100 or 1000
- ☐ simplify the fraction
- ☐ write a fraction with denominator 10, 100 or 1000 as a decimal

Key terms Write the definition as your teacher explains it.

Tenths

Hundredths

Thousandths

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write each decimal as a simplified fraction.

a Write 0.5 as a simplified fraction.

b Write 0.89 as a fraction.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Count the decimal places: 1 → tenths, 2 → hundredths, 3 → thousandths.

ROUND 1 Questions 1–16: write each decimal as a fraction in simplest form.

1 0.04

2 0.02

3 0.33

4 0.43

5 0.09

6 0.9

7 0.8

8 0.07

9 0.82

10 0.01

11 0.3

12 0.68

13 0.2

14 0.44

15 0.06

16 0.16

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: write each fraction as a decimal.

17 $\frac{47}{100}$

18 $\frac{35}{100}$

19 $\frac{96}{100}$

20 $\frac{1}{10}$

21 $\frac{86}{100}$

22 $\frac{4}{10}$

23 $\frac{24}{100}$

24 $\frac{89}{100}$

25 $\frac{56}{100}$

26 $\frac{15}{100}$

27 $\frac{98}{100}$

28 $\frac{188}{1000}$

29 $\frac{17}{1000}$

30 $\frac{404}{1000}$

31 $\frac{92}{100}$

32 $\frac{38}{100}$



Level 2 · Building up

Medium · 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write as a simplified fraction, then as a decimal.

a Write 0.444 as a simplified fraction.

Your teacher works this out. Copy every step.

b Write $\frac{4}{10}$ as a decimal.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Write the digits over 10, 100 or 1000.

ROUND 1

Questions 1–16: write as a fraction in simplest form.

1 0.708

2 0.927

3 0.376

4 0.555

5 0.661

6 0.286

7 0.487

8 0.818

9 0.855

10 0.619

11 0.963

12 0.951

13 0.157

14 0.277

15 0.826

16 0.759

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 17–22: write as a mixed numeral in simplest form.

17 3.37

18 2.75

19 3.55

20 4.25

21 3.66

22 2.82

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write each fraction as a decimal.

a Write $\frac{7}{100}$ as a decimal.

Your teacher works this out. Copy every step.

b Write $\frac{11}{1000}$ as a decimal.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Convert.

3 QUESTIONS

REMEMBER Simplify.

1 Write $\frac{3}{5}$ as a decimal (make the denominator 10).

2 Write $\frac{7}{20}$ as a decimal (make it 100).

3 Write $\frac{9}{250}$ as a decimal (make it 1000).

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 0.7 as a fraction.

Level 2

Write $\frac{23}{100}$ as a decimal.

Level 3

Write 0.35 as a simplified fraction.

Extension · Stretch yourself

Worked example

Write $\frac{17}{25}$ as a decimal.

1. $\times 4$ top and bottom: $\frac{68}{100}$

Answer: 0.68

YOU DO

On your own: read the worked example, then try these.

E1 Write $\frac{3}{40}$ as a decimal.

E2 Write 4.075 as a simplified mixed numeral.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Count the decimal places: 1 → tenths, 2 → hundredths, 3 → thousandths.
 2. Write the digits over 10, 100 or 1000.
 3. Simplify.
 4. To go back: make the denominator 10, 100 or 1000.

Level 1

0.5

1. $\frac{5}{10}$

Answer: $\frac{1}{2}$

Level 2

0.444

1. $\frac{444}{1000}$

Answer: $\frac{111}{250}$

Level 3

$\frac{11}{1000}$

1. 3 decimal places

Answer: 0.011

Extension

$\frac{17}{25}$

1. $\frac{68}{100}$

Answer: 0.68

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LESSON
7.03

Adding and subtracting decimals

Date _____

Learning intentions

We are learning to:

- add and subtract decimals

Success criteria

I can:

- ☐ line up the decimal points
- ☐ fill empty places with zeros
- ☐ estimate to check the answer

Key terms Write the definition as your teacher explains it.

Decimal point

Estimate

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Line up the decimal points.

a $12.35 + 4.53$

b $9.23 - 6.851$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Line up the decimal points.

ROUND 1 Questions 1–16: evaluate. Line up the decimal points.

1 $4.9 + 7.6$

2 $6.7 - 3.3$

3 $4.5 + 3.5$

4 $4.7 + 7.5$

5 $3.6 + 6.7$

6 $5.2 - 4.4$

7 $3.9 + 8.7$

8 $9.5 + 6.4$

9 $5.9 + 3.4$

10 $5.4 + 2.4$

11 $5.5 - 3.3$

12 $9.2 + 3.7$

13 $2.5 + 7.9$

14 $7.9 + 7$

15 $7.5 + 4.9$

16 $8.8 - 7.4$

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: evaluate.

17 $4.23 + 8.65$

18 $7.17 + 1.02$

19 $9.57 + 4.4$

20 $8.02 - 6.29$

21 $4.06 - 1.85$

22 $8.21 - 6.3$

23 $3.33 + 3.96$

24 $3.47 + 6.13$

25 $8.17 - 1.31$

26 $4.34 + 1.19$

27 $3.54 + 1.28$

28 $3.08 - 2$

29 $5.57 + 4.37$

30 $7.99 + 7.49$

31 $6.13 + 5.91$

32 $8.82 - 1.9$

Level 2 · Building up

Medium · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Evaluate. Fill gaps with zeros.

a $0.568 + 23 + 4.027$

Your teacher works this out. Copy every step.

b $214.33 - 109.84$

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Fill gaps with zeros.

ROUND 1 Questions 1–16: evaluate. Fill the empty places with zeros.

1 $29.5 + 5.82$

2 $182 - 2.54$

3 $30.9 + 4.09$

4 $28.8 - 9.5$

5 $13.6 + 3.8$

6 $35.3 + 5.81$

7 $9 + 9.02$

8 $31.4 + 6.82$

9 $121 + 8.56$

10 $30.4 + 9.61$

11 $117 - 7.59$

12 $363 + 6.41$

13 $320 + 1.99$

14 $20.1 - 9.94$

15 $11.7 - 6.14$

16 $145 + 5.1$

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 17–22: evaluate. Work from left to right.

17 $7.74 + 5.82 - 9.4$

18 $9.7 + 2.77 - 8.5$

19 $8.27 + 8.52 - 5.8$

20 $18.54 + 3.16 - 8.3$

21 $12.32 + 0.64 - 9.8$

22 $8.95 + 6.76 - 7.3$



Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Evaluate.

a $1453.6 + 1287.31 - 2344.4$

b $12.35 + 4.53 + 0.56 + 3.125 + 24.7 + 20.09$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Evaluate.

3 QUESTIONS

REMEMBER Add or subtract as for whole numbers.

1 $12.35 + 4.53 + 0.56 + 3.125 + 24.7$
 $+ 20.09$

② $1453.6 + 1287.31 - 2344.4$

3 Kim had \$50. She spent \$12.85 and \$7.40. How much is left?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
------	-------

Name _____ Class _____

 $2.7 + 1.6$ 
$$15.3 - 8.75$$

$$6.05 + 13.9 - 0.87$$

 Extension · Stretch yourself Worked example

Find the missing number: $4.8 - ? + 2.35 = 5$

- $4.8 + 2.35 = 7.15$
- $7.15 - ? = 5$

Answer: 2.15

YOU DO

On your own: read the worked example, then try these.

- E1 Find the missing number: $? + 3.07 - 1.5 = 10$.

- E2 Using each of the digits 1 to 6 once, make two decimals of the form $\square.\square\square$ with the largest possible sum.

Lesson summary

Missed the lesson? Start here.

Remember

- Line up the decimal points.
- Fill gaps with zeros.
- Add or subtract as for whole numbers.
- Put the point in line in the answer.

 Level 1

$$12.35 + 4.53$$

- Line up points

Answer: 16.88

 Level 2

$$9.23 - 6.851$$

- $9.230 - 6.851$

Answer: 2.379

 Level 3

$$214.33 - 109.84$$

- Borrow carefully

Answer: 104.49

 Extension

$$4.8 - ? + 2.35 = 5$$

- $7.15 - 5$

Answer: 2.15

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.04

Multiplying and dividing by powers of 10

Date _____

Learning intentions

We are learning to:

- multiply and divide decimals by 10, 100 and 1000

Success criteria

I can:

- ☐ move digits one place for each zero
- ☐ $\times$ moves digits left (bigger); $\div$ moves digits right (smaller)
- ☐ add zeros as place holders

Key terms Write the definition as your teacher explains it.

Power of 10

Place holder

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Multiply by 10 and 100.

a 7.54×10

b 7.54×100

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Count the zeros in 10, 100, 1000.

ROUND 1 Questions 1–16: multiply. Move the digits left by one place for each zero.

1 41.2×100

2 5.95×100

3 0.66×100

4 0.655×10

5 91×100

6 $92.3 \times 1\ 000$

7 $0.831 \times 1\ 000$

8 93.2×10

9 $49.3 \times 1\ 000$

10 $43.1 \times 1\ 000$

11 24.9×100

12 0.908×10

13 0.448×10

14 0.183×100

15 29.5×10

16 $76.4 \times 1\ 000$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: find the missing power of 10.

17 $5.36 \div \square = 0.536$

18 $2.48 \times \square = 24.8$

19 $1.61 \times \square = 1\,610$

20 $4.25 \times \square = 425$

21 $6.39 \times \square = 63.9$

22 $6.18 \div \square = 0.618$

23 $2 \times \square = 20$

24 $3.95 \times \square = 3\,950$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Divide by 100 and 1000.

a $13.9 \div 100$

b $13.9 \div 1000$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER $\div$: move the digits that many places right.

1 A pen costs \$1.35. Find the cost of 100 pens.

2 Ten friends share a \$87.50 bill. How much each?

3 1000 sheets of paper stack 96 mm high. How thick is one sheet?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

3.8×10

Level 2

$52.6 \div 100$

Level 3

0.049×1000

 Extension · Stretch yourself Worked example

Evaluate 0.025×400 without a calculator.

1. $400 = 4 \times 100$
2. $0.025 \times 100 = 2.5$
3. 2.5×4

Answer: 10

YOU DO

On your own: read the worked example, then try these.

- E1 Evaluate 3.6×2000 .

- E2 Evaluate $48 \div 300$ (hint: $\div 3$ then $\div 100$).

Lesson summary

Missed the lesson? Start here.

Remember

1. Count the zeros in 10, 100, 1000.
3. $\div$: move the digits that many places right.

2. $\times$: move the digits that many places left.
4. Use zeros to fill empty places.

 Level 1

$$7.54 \times 100$$

1. 2 places left

Answer: 754

 Level 2

$$13.9 \div 100$$

1. 2 places right

Answer: 0.139

 Level 3

$$0.0072 \times 1000$$

1. 3 places left

Answer: 7.2

 Extension

$$0.025 \times 400$$

1. 2.5×4

Answer: 10

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LESSON
7.05

Using a known product

Date _____

Learning intentions

We are learning to:

- use a known product to multiply decimals

Success criteria

I can:

- ☐ count the decimal places in the question
- ☐ put the same number of decimal places in the answer
- ☐ estimate to check the size of the answer

Key terms Write the definition as your teacher explains it.**Product****Decimal places**

Notes

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Use $42 \times 76 = 3192$.**a** Given $42 \times 76 = 3192$, find 4.2×76 .**b** Find 4.2×7.6 .

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS****REMEMBER** Ignore the decimal points and use the known product.**ROUND 1** Questions 1–8: use the fact given to find the answer. Count the decimal places.**1** $36 \times 27 = 972$. Find 0.36×2.7 .**2** $31 \times 38 = 1\,178$. Find 0.31×0.38 .**3** $46 \times 22 = 1\,012$. Find 0.46×22 .**4** $29 \times 28 = 812$. Find 2.9×2.8 .**5** $41 \times 29 = 1\,189$. Find 4.1×0.29 .**6** $23 \times 28 = 644$. Find 2.3×28 .**7** $18 \times 33 = 594$. Find 0.18×3.3 .**8** $18 \times 19 = 342$. Find 0.18×0.19 .

YOU DO On your own: **Medium** keep going: Round 2.

4 QUESTIONS MORE

ROUND 2 Questions 9–12: use the fact given to find each answer.

9 $17 \times 26 = 442$. Find $1\,700 \times 2.6$.

10 $49 \times 25 = 1\,225$. Find $4\,900 \times 2.5$.

11 $33 \times 20 = 660$. Find $3\,300 \times 0.2$.

12 $31 \times 22 = 682$. Find $3\,100 \times 0.22$.

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use $42 \times 76 = 3\,192$.

a Find 42×7.6 .

b Find 0.42×0.76 .

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Given $125 \times 36 = 4\,500$, find:

3 QUESTIONS

REMEMBER The answer has the same total.

1 1.25×36

2 12.5×3.6

3 0.125×0.36

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Given $23 \times 14 = 322$, find 2.3×14 .

Level 2

Find 2.3×1.4 .

Level 3

Find 0.23×0.14 .

 Extension · Stretch yourself Worked example

Given $17 \times 24 = 408$, find $408 \div 1.7$.

- $408 \div 17 = 24$
- Dividing by a number 10 times smaller gives an answer 10 times bigger

Answer: 240

YOU DO

On your own: read the worked example, then try these.

E1 Given $17 \times 24 = 408$, find $40.8 \div 2.4$.

E2 Given $17 \times 24 = 408$, find $0.408 \div 0.017$.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Ignore the decimal points and use the known product.
 - Count the decimal places in the question.
 - The answer has the same total.
 - Estimate to check.

 Level 1

$$4.2 \times 76$$

- 1 decimal place

Answer: 319.2

 Level 2

$$4.2 \times 7.6$$

- 2 decimal places

Answer: 31.92

 Level 3

$$0.42 \times 0.76$$

- 4 decimal places

Answer: 0.3192

 Extension

$$408 \div 1.7$$

- 10×24

Answer: 240

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.06

Multiplying decimals

Date _____

Learning intentions

We are learning to:

- multiply decimals

Success criteria

I can:

- ☐ multiply as whole numbers first
- ☐ count the total decimal places in the question
- ☐ place the decimal point in the answer

Key terms Write the definition as your teacher explains it.

Product

Decimal places

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Count the decimal places.

a 2.75×6

b 0.5×1.2

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Multiply the numbers ignoring the points.

ROUND 1 Questions 1–16: evaluate.

1 6.5×4

2 8.3×7

3 1.2×9

4 2.7×2

5 9.4×7

6 2.1×4

7 7.5×8

8 1.6×7

9 7.6×8

10 3.8×2

11 5.1×5

12 1.2×5

13 8.1×5

14 4.1×2

15 6.9×3

16 6.9×7

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate. Count the decimal places in the question.

17 0.9×0.2

18 0.7×0.05

19 0.3×0.06

20 0.2×0.05

21 0.3×0.2

22 0.3×0.09

23 0.2×0.4

24 0.5×0.03

25 0.4×0.3

26 0.5×0.2

27 0.2×0.01

28 0.3×0.9

29 0.7×0.01

30 0.5×0.6

31 0.3×0.5

32 0.1×0.09



Level 2 • Building up

Medium • 19 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a 6.1×1.2

Your teacher works this out. Copy every step.

b 3.25×0.41

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line (show working in the boxes).

19 QUESTIONS

REMEMBER Count all decimal places in the question.

ROUND 1

Questions 1–16: evaluate.

1 0.12×6.2

2 0.32×5

3 0.83×5.9

4 0.48×4.9

5 0.78×3.4

6 0.33×4.8

7 0.74×6

8 0.43×5.6

9 0.46×1.7

10 0.26×9.4

11 0.73×8.4

12 0.6×4.9

13 0.31×5.9

14 0.95×2.6

15 0.61×3.3

16 0.23×6.9

YOU DO On your own: **Medium** keep going: Round 2.

3 QUESTIONS MORE

ROUND 2 Questions 17–19: evaluate on paper.

17 4.16×3.9

18 2.11×3.9

19 3.35×7.9

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate.

a 0.05×0.02

Your teacher works this out. Copy every step.

b 4.67×1.1

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Give the answer that many decimal places.

- 1 Find the area of a rectangle 4.5 m by 3.2 m.

- 2 Petrol costs \$1.89 per litre. Find the cost of 42.5 L (to the nearest cent).

- 3 Find 0.2^3 .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

4.3×2

Level 2

0.6×0.7

Level 3

2.35×1.4

 Extension · Stretch yourself Worked exampleEvaluate $1.2 \times 1.2 \times 1.2$.

- $1.2 \times 1.2 = 1.44$
- $1.44 \times 1.2 = 1.728$

Answer: 1.728

YOU DO

On your own: read the worked example, then try these.

- E1 Evaluate $0.3 \times 0.3 \times 0.3 \times 0.3$.

- E2 Find two decimals whose product is 0.24.

Lesson summary

Missed the lesson? Start here.

Remember

- Multiply the numbers ignoring the points.
- Count all decimal places in the question.
- Give the answer that many decimal places.
- Estimate: $6.1 \times 1.2 \approx 6 \times 1 = 6$.

 Level 1 2.75×6

- $2.75 \times 6 = 16.50$
- 2 decimal places

Answer: 16.5

 Level 2 0.05×0.02

- $5 \times 2 = 10$
- 4 decimal places

Answer: 0.001

 Level 3 3.25×0.41

- $325 \times 41 = 13\,325$
- 4 places

Answer: 1.3325

 Extension 1.2^3

- 1.44×1.2

Answer: 1.728

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.07

Dividing decimals by whole numbers

Date _____

Learning intentions

We are learning to:

- divide a decimal by a whole number

Success criteria

I can:

- ☐ use short division, keeping the decimal point in line
- ☐ add zeros to keep dividing
- ☐ solve sharing problems

Key terms Write the definition as your teacher explains it.

Quotient

Short division

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate using short division.

a $762.4 \div 2$

b $97.6 \div 8$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **32 QUESTIONS**

REMEMBER Set out short division.

ROUND 1 Questions 1–16: evaluate. Keep the decimal point in line.

1 $18 \div 2$

2 $19.2 \div 2$

3 $26.6 \div 7$

4 $24.9 \div 3$

5 $11.2 \div 2$

6 $13.5 \div 9$

7 $39.2 \div 8$

8 $12 \div 2$

9 $34.8 \div 4$

10 $8.6 \div 2$

11 $55.3 \div 7$

12 $10.8 \div 2$

13 $69.6 \div 8$

14 $17.8 \div 2$

15 $4.4 \div 4$

16 $34.4 \div 4$

YOU DO On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2 Questions 17–32: evaluate.

17 $32.56 \div 4$

18 $17.65 \div 5$

19 $5.65 \div 5$

20 $59.78 \div 7$

21 $2.94 \div 2$

22 $52.16 \div 8$

23 $35.8 \div 5$

24 $48.18 \div 6$

25 $88.92 \div 9$

26 $17.72 \div 4$

27 $55.12 \div 8$

28 $78.16 \div 8$

29 $10.56 \div 6$

30 $22.56 \div 8$

31 $7.24 \div 2$

32 $21.69 \div 3$

Level 2 · Building up

Medium · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Evaluate. Keep the decimal point in line.

a $195.6 \div 3$

Your teacher works this out. Copy every step.

b $2.75 \div 4$

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Put the decimal point in the answer above the point in the question.

ROUND 1 Questions 1–16: evaluate. Add zeros after the last digit if you need to.

1 $4 \div 4$

2 $2.9 \div 5$

3 $4.8 \div 2$

4 $2 \div 4$

5 $3.1 \div 8$

6 $8.4 \div 8$

7 $7 \div 4$

8 $4.8 \div 4$

9 $6.6 \div 4$

10 $8.6 \div 2$

11 $3.9 \div 8$

12 $3.8 \div 8$

13 $5 \div 8$

14 $4.2 \div 4$

15 $2.8 \div 2$

16 $9.4 \div 2$

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 17–22: share the amount equally.

17 \$44.80 between 7 people

18 \$7.40 between 5 people

19 \$60.24 between 4 people

20 \$79.12 between 4 people

21 \$124.25 between 5 people

22 \$166.40 between 8 people



Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Evaluate. Add zeros if needed.

a $3.5 \div 8$

b Share \$50.40 among 6 people.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Divide each digit, carrying remainders.

1 Share \$50.40 among 6 people.

2 A 4.5 m plank is cut into 4 equal pieces. How long is each?

3 8 cans of drink cost \$11.60. Find the cost of one can.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
John	Math
Jane	Math
John	Science
Jane	Science
John	History
Jane	History
John	Art
Jane	Art
John	Music
Jane	Music
John	Physical Education
Jane	Physical Education



$6.8 \div 2$



$14.4 \div 6$



$5.3 \div 4$

 Extension · Stretch yourself Worked example

The mean of 3 decimals is 2.45. Two of them are 1.8 and 3.1. Find the third.

1. Total = $3 \times 2.45 = 7.35$
2. $7.35 - 1.8 - 3.1$

Answer: 2.45

YOU DO

On your own: read the worked example, then try these.

- E1 The mean of 4 numbers is 5.6. Three are 4.2, 7.1 and 5.5. Find the fourth.

- E2 Find $10 \div 8$ and $10 \div 16$. What do you notice?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Set out short division.
 2. Put the decimal point in the answer above the point in the question.
 3. Divide each digit, carrying remainders.
 4. Add zeros after the point to keep going.

 Level 1

$$762.4 \div 2$$

1. Divide each digit

Answer: 381.2

 Level 2

$$97.6 \div 8$$

1. $9 \div 8 = 1$ r1, $17 \div 8 = 2$ r1, $16 \div 8 = 2$

Answer: 12.2

 Level 3

$$2.75 \div 4$$

1. $2.750 \div 4$

Answer: 0.6875

 Extension

Mean 2.45, find third

1. $7.35 - 4.9$

Answer: 2.45

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.08

Dividing by a decimal

Date _____

Learning intentions

We are learning to:

- divide by a decimal

Success criteria

I can:

- ☐ multiply both numbers by 10, 100 or 1000 to make the divisor whole
- ☐ then divide by the whole number
- ☐ check that the answer makes sense

Key terms Write the definition as your teacher explains it.**Divisor****Equivalent division**

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate. Make the divisor a whole number.

a $12.5 \div 0.5$

b $12.72 \div 0.4$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DOOn your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER Make the divisor a whole number.**ROUND 1** Questions 1–16: evaluate. Make the divisor a whole number first ($\times 10$ both).

1 $16 \div 0.8$

2 $4.4 \div 0.4$

3 $4.8 \div 0.2$

4 $8.7 \div 0.3$

5 $1.8 \div 0.2$

6 $9.9 \div 0.9$

7 $4 \div 0.5$

8 $26.1 \div 0.9$

9 $8.8 \div 0.4$

10 $12 \div 0.5$

11 $4 \div 0.8$

12 $0.9 \div 0.3$

13 $3.2 \div 0.4$

14 $1.4 \div 0.7$

15 $6 \div 0.4$

16 $4.2 \div 0.7$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: evaluate.

17 $20 \div 0.5$

18 $2.2 \div 0.1$

19 $4.2 \div 0.2$

20 $7 \div 0.2$

21 $19 \div 0.5$

22 $3.2 \div 0.1$

23 $2 \div 0.5$

24 $0.75 \div 0.25$

25 $5.2 \div 0.2$

26 $3 \div 0.2$

27 $3.8 \div 0.2$

28 $16.5 \div 0.5$

29 $0.5 \div 0.1$

30 $3.3 \div 0.1$

31 $9.75 \div 0.25$

32 $1.4 \div 0.2$



Level 2 • Building up

Medium • 22 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $6.9 \div 0.03$

Your teacher works this out. Copy every step.

b $0.508 \div 0.02$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

22 QUESTIONS

REMEMBER Multiply both numbers by the same power of 10.

ROUND 1

Questions 1–16: evaluate.

1 $7.11 \div 0.9$

2 $0.88 \div 0.2$

3 $3.2 \div 0.4$

4 $0.68 \div 0.2$

5 $0.6 \div 0.3$

6 $5.46 \div 0.7$

7 $1.44 \div 0.6$

8 $1.34 \div 0.2$

9 $5.36 \div 0.8$

10 $5.04 \div 0.9$

11 $2.25 \div 0.9$

12 $0.33 \div 0.3$

13 $2.73 \div 0.7$

14 $4.25 \div 0.5$

15 $3.96 \div 0.9$

16 $1.38 \div 0.2$

YOU DO

On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 17–22: evaluate. Multiply both numbers by 100 first.

17 $4.44 \div 0.06$

18 $5.53 \div 0.07$

19 $6.75 \div 0.09$

20 $0.45 \div 0.09$

21 $3.12 \div 0.06$

22 $3.04 \div 0.04$



Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Evaluate, then solve the problem.

a $4 \div 0.8$

Your teacher works this out. Copy every step.

b How many 0.25 L cups can be filled from 3.5 L?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Problems.

3 QUESTIONS

REMEMBER Then divide as usual.

1 How many 0.25 L cups can be filled from 3.5 L?

2 A car travels 157.5 km on 12.6 L of fuel. How many km per litre?

3 Ribbon costs \$0.45 per metre.
How many metres for \$6.30?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name	Class
------	-------



$8 \div 0.4$



$5.4 \div 0.6$



$0,36 \div 0,09$

Extension · Stretch yourself

Worked example

Evaluate $0.0144 \div 0.012$.

1. $\times 1000$: $14.4 \div 12$

Answer: 1.2

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $3.24 \div 0.018$.

E2 Is $8 \div 0.1$ bigger or smaller than 8×10 ? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Make the divisor a whole number.
 2. Multiply both numbers by the same power of 10.
 3. Then divide as usual.
 4. Dividing by a number less than 1 makes the answer bigger.

Level 1

$12.5 \div 0.5$

1. $125 \div 5$

Answer: 25

Level 2

$12.72 \div 0.4$

1. $127.2 \div 4$

Answer: 31.8

Level 3

$6.9 \div 0.03$

1. $690 \div 3$

Answer: 230

Extension

$0.0144 \div 0.012$

1. $14.4 \div 12$

Answer: 1.2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.09

Fractions to decimals

Date _____

Learning intentions

We are learning to:

- convert fractions to decimals, including recurring decimals

Success criteria

I can:

- ☐ divide the numerator by the denominator
- ☐ recognise a **recurring** decimal
- ☐ write recurring decimals using dot notation

Key terms Write the definition as your teacher explains it.

Terminating decimal

Recurring decimal

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a decimal.

a $\frac{4}{5}$ as a decimal

b $\frac{3}{8}$ as a decimal

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER Divide the top by the bottom.

ROUND 1 Questions 1–16: write each fraction as a decimal.

1 $\frac{1}{8}$

2 $\frac{1}{4}$

3 $\frac{1}{2}$

4 $\frac{7}{8}$

5 $\frac{11}{20}$

6 $\frac{3}{10}$

7 $\frac{1}{5}$

8 $\frac{1}{50}$

9 $\frac{5}{8}$

10 $\frac{1}{25}$

11 $\frac{3}{5}$

12 $\frac{9}{10}$

13 $\frac{2}{5}$

14 $\frac{19}{20}$

15 $\frac{17}{20}$

16 $\frac{17}{25}$

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: write as a recurring decimal. Put a dot over the digit that repeats.

17 $\frac{5}{6}$

18 $\frac{7}{6}$

19 $\frac{1}{12}$

20 $\frac{1}{15}$

21 $\frac{1}{3}$

22 $\frac{2}{3}$

23 $\frac{5}{3}$

24 $\frac{8}{9}$

25 $\frac{1}{9}$

26 $\frac{2}{9}$

27 $\frac{1}{6}$

28 $\frac{4}{9}$

29 $\frac{5}{9}$

30 $\frac{7}{9}$

31 $\frac{4}{3}$

32 $\frac{1}{30}$



Level 2 · Building up

Medium · 24 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write as a decimal. Use dot notation.

a $\frac{2}{3}$ as a decimal

Your teacher works this out. Copy every step.

b $\frac{5}{9}$ as a decimal

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Add zeros after the decimal point to keep going.

ROUND 1

Questions 1–16: write each mixed numeral as a decimal.

1 $2\frac{2}{5}$

2 $7\frac{4}{5}$

3 $2\frac{22}{25}$

4 $7\frac{12}{25}$

5 $2\frac{3}{8}$

6 $4\frac{1}{20}$

7 $3\frac{5}{8}$

8 $1\frac{5}{8}$

9 $3\frac{1}{2}$

10 $6\frac{24}{25}$

11 $4\frac{2}{5}$

12 $5\frac{1}{2}$

13 $4\frac{1}{8}$

14 $4\frac{1}{2}$

15 $9\frac{18}{25}$

16 $6\frac{1}{5}$

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: which is larger? Change the fraction to a decimal first.

17 $\frac{3}{5}$ or 0.01

18 $\frac{1}{20}$ or 0.23

19 $\frac{4}{5}$ or 0.21

20 $\frac{3}{5}$ or 0.78

21 $\frac{3}{4}$ or 0.84

22 $\frac{11}{20}$ or 0.6

23 $\frac{1}{4}$ or 0.53

24 $\frac{1}{4}$ or 0.78

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a recurring decimal.

a $\frac{5}{11}$ as a decimal

Your teacher works this out. Copy every step.

b $\frac{1}{6}$ as a decimal

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Terminating: the division ends.

1 Order $\frac{2}{3}$, 0.6, 0.67, $\frac{5}{8}$ from smallest to largest.

2 Write $\frac{1}{9}$, $\frac{2}{9}$ and $\frac{3}{9}$ as decimals. Predict $\frac{7}{9}$.

3 Write $\frac{1}{7}$ as a decimal. How many digits repeat?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write $\frac{3}{5}$ as a decimal.

Level 2

Write $\frac{7}{8}$ as a decimal.

Level 3

Write $\frac{1}{3}$ as a recurring decimal.

 Extension · Stretch yourself Worked example

Write $0.\dot{4}$ as a fraction.

1. $0.444... = 4 \times 0.111...$

2. $0.111... = \frac{1}{9}$

Answer: $\frac{4}{9}$

YOU DO

On your own: read the worked example, then try these.

E1 Write $0.\dot{7}$ as a fraction.

E2 Write $0.\dot{1}\dot{8}$ ($0.181\ 818...$) as a fraction. Hint: $0.\dot{0}\dot{1} = \frac{1}{99}$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Divide the top by the bottom.
3. Terminating: the division ends.

2. Add zeros after the decimal point to keep going.
4. Recurring: digits repeat forever. Show them with a dot: $0.\dot{6}$.

 Level 1

$$\frac{4}{5}$$

1. $4 \div 5$

Answer: 0.8

 Level 2

$$\frac{3}{8}$$

1. $3.000 \div 8$

Answer: 0.375

 Level 3

$$\frac{2}{3}$$

1. $2 \div 3 = 0.666...$

Answer: $0.\dot{6}$

 Extension

$$0.\dot{4}$$

1. 4 ninths

Answer: $\frac{4}{9}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.10

Rounding decimals

Date _____

Learning intentions

We are learning to:

- round decimals to a given number of decimal places

Success criteria

I can:

- ☐ find the last digit to keep
- ☐ look at the next digit: 5 or more rounds up
- ☐ round to the nearest tenth, hundredth or thousandth

Key terms Write the definition as your teacher explains it.

Decimal places

Nearest tenth

Notes

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Round to the place given.

a Round 406.28 to the nearest tenth.

b Round 125.724 to 2 decimal places.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER Find the last digit you keep.

ROUND 1 Questions 1–16: round to 1 decimal place.

1 38.82

2 12.53

3 70.56

4 17.82

5 23.39

6 31.79

7 74.77

8 86.69

9 9.15

10 26.02

11 3.29

12 87.81

13 1.07

14 38.34

15 49.45

16 70.07

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 17–32: round to 2 decimal places.

17 81.846

18 18

19 4.664

20 25.334

21 11.97

22 90.53

23 44.742

24 69.055

25 70.736

26 87.015

27 12.883

28 83.195

29 73.451

30 17.697

31 95.801

32 35.486



Level 2 · Building up

Medium · 24 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Round to the place given.

a Round 3.8967 to the nearest hundredth.

Your teacher works this out. Copy every step.

b Round 2345.876 to 1 decimal place.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Look at the next digit to the right.

ROUND 1

Questions 1–16: round to the place given.

1 25.2318 (nearest whole)

2 36.6978 (3 d.p.)

3 31.5361 (3 d.p.)

4 96.5978 (nearest whole)

5 89.871 (2 d.p.)

6 64.8817 (2 d.p.)

7 99.9926 (1 d.p.)

8 80.9139 (nearest whole)

9 39.9886 (3 d.p.)

10 60.5283 (1 d.p.)

11 28.5394 (3 d.p.)

12 71.9166 (nearest whole)

13 20.3857 (1 d.p.)

14 72.511 (nearest whole)

15 89.8289 (3 d.p.)

16 43.3307 (3 d.p.)

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: use your calculator, then round the answer to 2 decimal places.

17 $27 \div 4$

18 $38 \div 3$

19 $35 \div 6$

20 $40 \div 11$

21 $31 \div 12$

22 $43 \div 13$

23 $20 \div 12$

24 $30 \div 12$

Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Round to the place given.

a Round 78 654.056 to 2 decimal places.

Your teacher works this out. Copy every step.

b Round 678.4309 to the nearest thousandth.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Rounding in context.

3 QUESTIONS

REMEMBER 5 or more: round up. 4 or less: leave.

1 Write $2 \div 3$ to 2 decimal places.

2 A bill of \$47.23 is shared by 3 people. How much each, to the nearest cent?

3 A number rounds to 4.7 (1 d.p.). What are the smallest and largest it could be (2 d.p.)?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Round 5.67 to 1 decimal place.

Level 2

Round 12.345 to 2 decimal places.

Level 3

Round 0.8996 to the nearest thousandth.

 Extension · Stretch yourself Worked example

Round 0.049 95 to 3 significant figures.

1. Significant figures start at the first non-zero digit: 4
2. Keep 4, 9, 9; next digit is 5, round up

Answer: 0.0500

YOU DO

On your own: read the worked example, then try these.

- E1** Round 23 461 to 2 significant figures.

- E2** Round 0.006 074 to 2 significant figures.

Lesson summary

Missed the lesson? Start here.

Remember

1. Find the last digit you keep.
2. Look at the next digit to the right.
3. 5 or more: round up. 4 or less: leave.
4. Drop the digits after.

 Level 1

406.28 to 1 d.p.

1. Next digit 8

Answer: 406.3

 Level 2

3.8967 to 2 d.p.

1. Next digit 6: $3.89 \rightarrow 3.90$

Answer: 3.90

 Level 3

678.4309 to 3 d.p.

1. Next digit 9

Answer: 678.431

 Extension

0.049 95 to 3 s.f.

1. Round up

Answer: 0.0500

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LESSON
7.11

Decimals and money

Date _____

Learning intentions

We are learning to:

- solve money problems with decimals

Success criteria

I can:

- ☐ choose the operations the problem needs
- ☐ set out working clearly with \$ and 2 decimal places
- ☐ round money to the nearest cent or 5 cents

Key terms Write the definition as your teacher explains it.

Total

Change

Notes

Level 1 · Getting started

Easy · 25 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the total or the price of one.

- a** A club orders 16 pizzas at \$19.70 each and 10 at \$15.50 each. Find the total cost.

- b** Riley bought 825 bricks for \$1105.50. How much did one brick cost?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **25 QUESTIONS**

REMEMBER Read carefully and list what you know.

ROUND 1 Questions 1–9: add the prices.

1 $\$1.67 + \10.03

2 $\$5.40 + \4.35

3 $\$0.68 + \8.93

4 $\$5.09 + \17.46

5 $\$13.18 + \16.72

6 $\$5.15 + \1.14

7 $\$1.78 + \18.95

8 $\$4.21 + \15.01

9 $\$6.15 + \15.93

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: find the change.

10 \$7.71 from \$20

11 \$4.21 from \$10

12 \$13.58 from \$50

13 \$8.55 from \$10

14 \$32.14 from \$50

15 \$4.61 from \$50

16 \$19.24 from \$50

17 \$9.21 from \$10

18 \$24.64 from \$50

19 \$9.95 from \$10

20 \$21.27 from \$50

21 \$20.75 from \$50

22 \$47.43 from \$50

23 \$10.72 from \$20

24 \$4.24 from \$10

25 \$19.63 from \$20

Level 2 · Building up

Medium · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find how much is left, then how much she must borrow.

- a** Mia saved \$220. She spent \$18.80, \$116.75, \$25 and \$37.70. How much is left?

Your teacher works this out. Copy every step.

- b** Mia wants a \$59.95 backpack. How much must she borrow?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS**

REMEMBER Choose +, −, × or ÷ for each step.

ROUND 1 Questions 1–12: find the total cost.

- 1** 3 items at \$12.15 each

- 2** 5 items at \$8.80 each

- 3** 5 items at \$12.01 each

- 4** 2 items at \$8.04 each

- 5** 5 items at \$2.18 each

- 6** 9 items at \$4.80 each

- 7** 7 items at \$4.98 each

- 8** 6 items at \$1.14 each

- 9** 5 items at \$4.28 each

- 10** 7 items at \$11.11 each

- 11** 6 items at \$11.53 each

- 12** 4 items at \$5.62 each

YOU DO

On your own: **Medium** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: round to the nearest 5 cents (cash rounding).

13 \$70.96

14 \$93.48

15 \$60.68

16 \$93.79

17 \$81.08

18 \$89.13

19 \$32.75

20 \$9.92

21 \$51.31

22 \$75.11

23 \$55.85

24 \$77.19

25 \$86.25

26 \$85.67

27 \$18.28

28 \$57.36



Level 3 • Challenge

Challenging • 3 questions • then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Round money, then find the change.

a Round \$17.83 to the nearest 5 cents.

Your teacher works this out. Copy every step.

b Find the change from \$50 for items costing \$12.95, \$8.40 and \$19.99.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Multi-step problems.

3 QUESTIONS

REMEMBER Write money with \$ and 2 decimal places.

1 Mia saved \$220. She spent \$18.80 on fares, \$116.75 on a ticket, \$25 on a program and \$37.70 on food. How much must she borrow to buy a \$59.95 backpack?

2 A phone plan costs \$35 a month plus \$0.15 per MB over the limit. Last month Sam went 64 MB over. Find the bill.

3 Which is the better buy: 1.5 L for \$3.60 or 2 L for \$4.60?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A shop sells pens at 3 for \$4.20 or 5 for \$6.50.
Which is cheaper per pen?

1. $4.20 \div 3 = \$1.40$

2. $6.50 \div 5 = \$1.30$

Answer: 5 for \$6.50

YOU DO

On your own: read the worked example, then try these.

- E1** You need at least 7 pens. Find the cheapest way to buy them using the packs.

- E2** A pack of 12 costs \$14.40. Is it better than 5 for \$6.50?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find $4 \times \$3.65$.

Level 2

Find the change from \$20 for \$7.45 and \$6.90.

Level 3

Six tickets cost \$76.50. Find the cost of one ticket.

Lesson summary

Missed the lesson? Start here.

Remember

1. Read carefully and list what you know.
2. Choose +, −, × or ÷ for each step.
3. Write money with \$ and 2 decimal places.
4. Check the answer is sensible.

Level 1

Pizza order

1. $16 \times 19.70 = 315.20$
2. $10 \times 15.50 = 155.00$

Answer: \$470.20

Level 2

One brick

1. $1105.50 \div 825$

Answer: \$1.34

Level 3

Mia borrows

1. $220 - 198.25 = 21.75$
2. $59.95 - 21.75$

Answer: \$38.20

Extension

Pens

1. \$1.40 vs \$1.30

Answer: 5 for \$6.50

My reflection: what did I learn today? Which question was hardest, and why?

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